

Understanding The

Cryptocurrency Analysis Using Machine Learning

An Insight into Crypto and Equity Trends
through a Data-Driven Lens



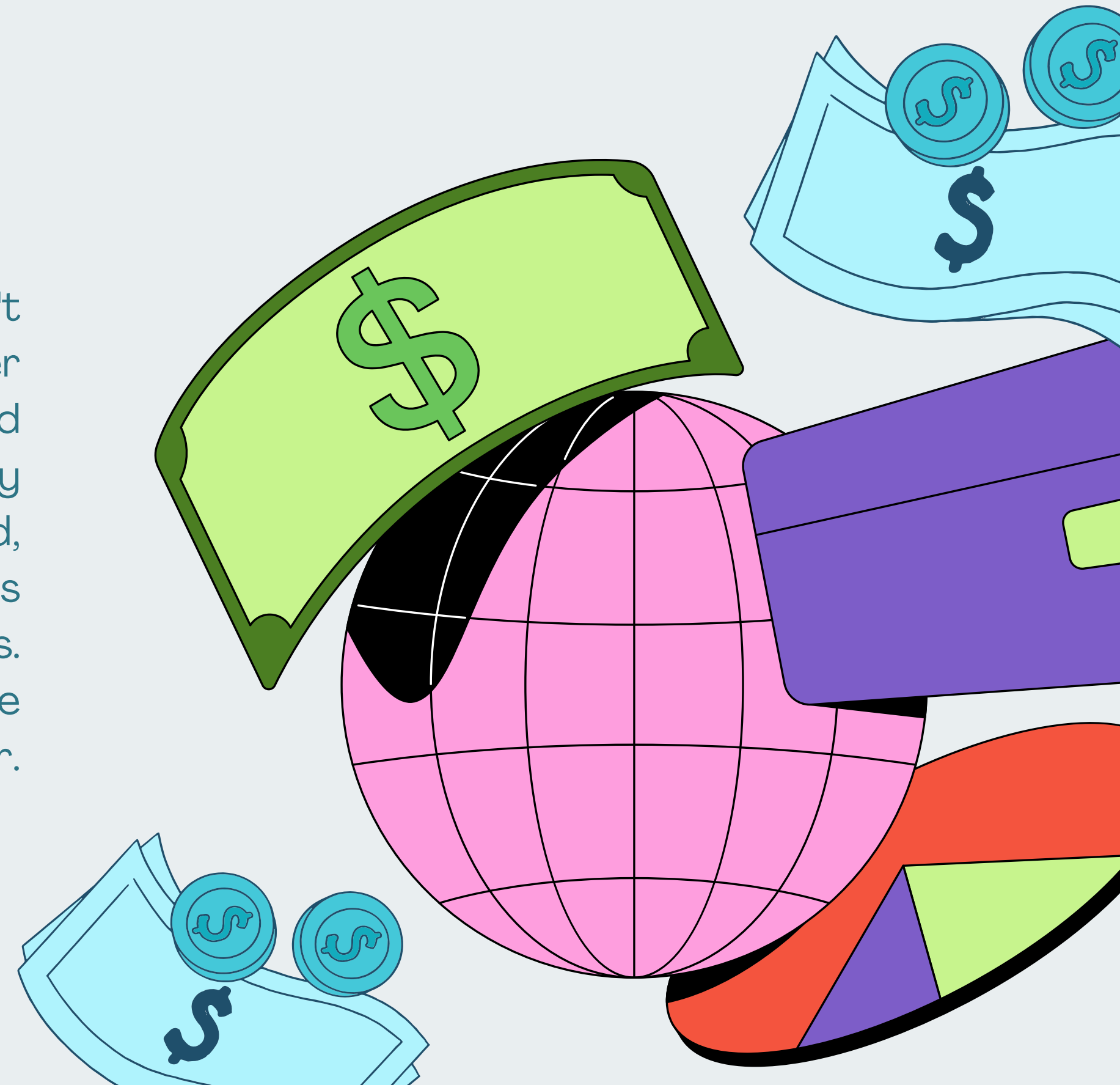
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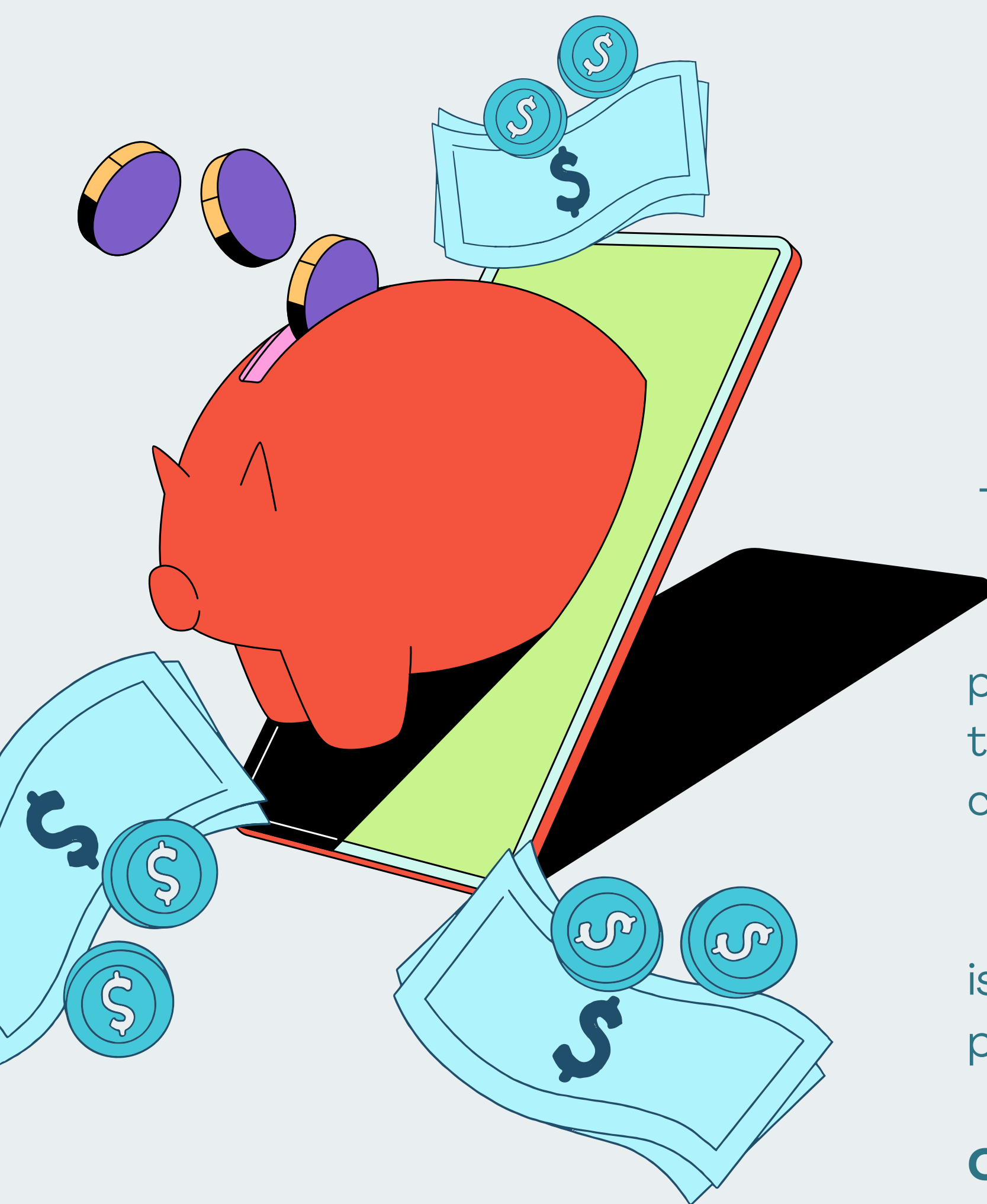
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Artificial Intelligence & Data Science

What is Cryptocurrency

Cryptocurrency is a digital payment system that doesn't rely on banks to verify transactions. It's a peer-to-peer system that can enable anyone anywhere to send and receive payments. Instead of being physical money carried around and exchanged in the real world, cryptocurrency payments exist purely as digital entries to an online database describing specific transactions. When you transfer cryptocurrency funds, the transactions are recorded in a public ledger. Cryptocurrency is stored in digital wallets





How to Invest in Cryptocurrency

There are two main ways to get cryptocurrency:

1. Mining:

Units of cryptocurrency are created through a process called mining, which involves using computer power to solve complicated mathematical problems that generate coins.

2. Buying from Brokers or Exchanges:

The most common way to invest in cryptocurrency is to buy it from a crypto exchange or broker. Some popular exchanges include Coinbase, Binance, and WazirX.

Once you buy cryptocurrency, you store it in a digital wallet.

Risks of Investing in Cryptocurrency

- Cryptocurrency payments do not come with legal protections.
- Cryptocurrency payments typically are not reversible.
- A cryptocurrency's value can change constantly and dramatically. An investment that may be worth thousands of dollars today could be worth only hundreds tomorrow. If the value goes down, there's no guarantee that it will rise again



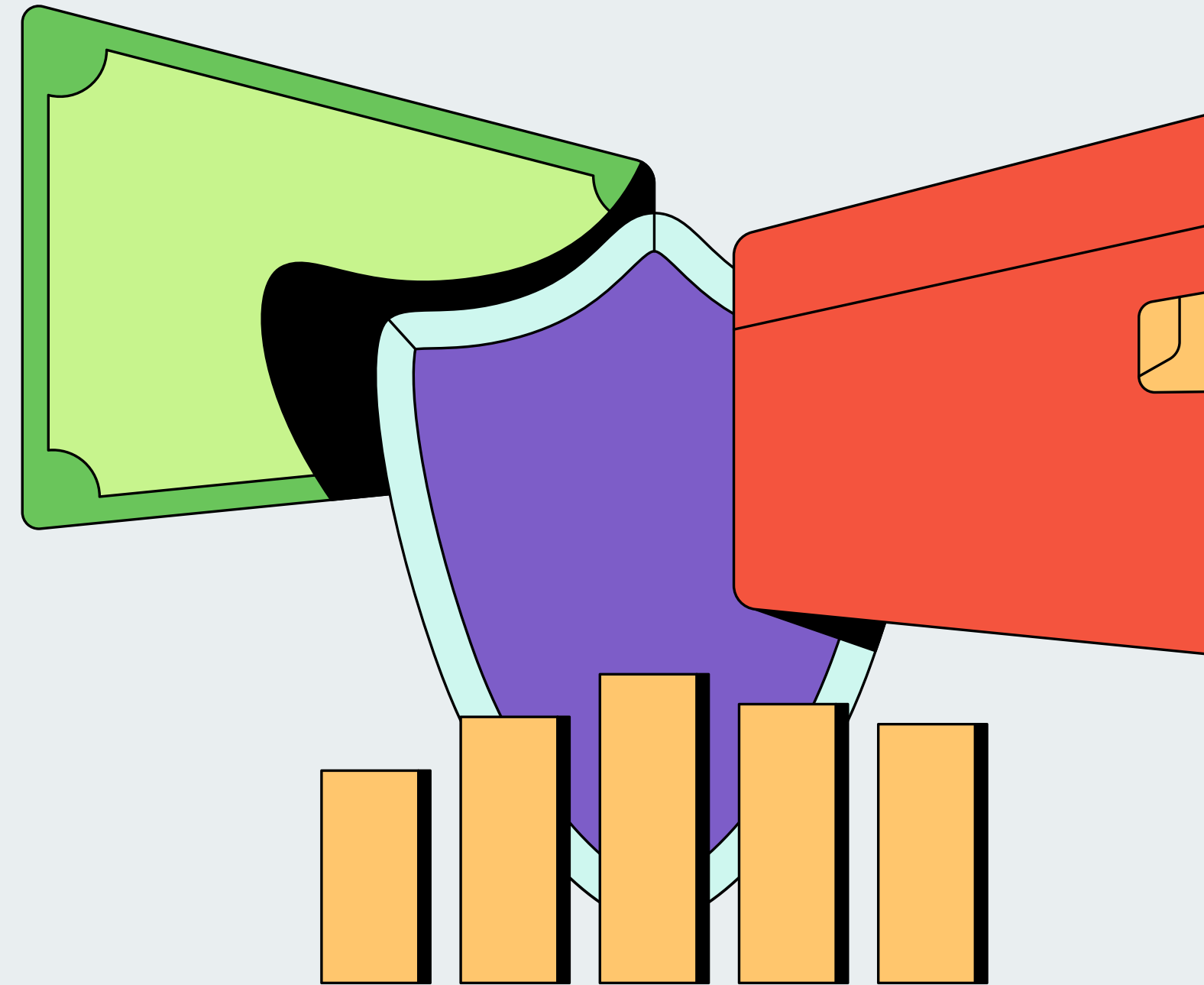
What is Bitcoin



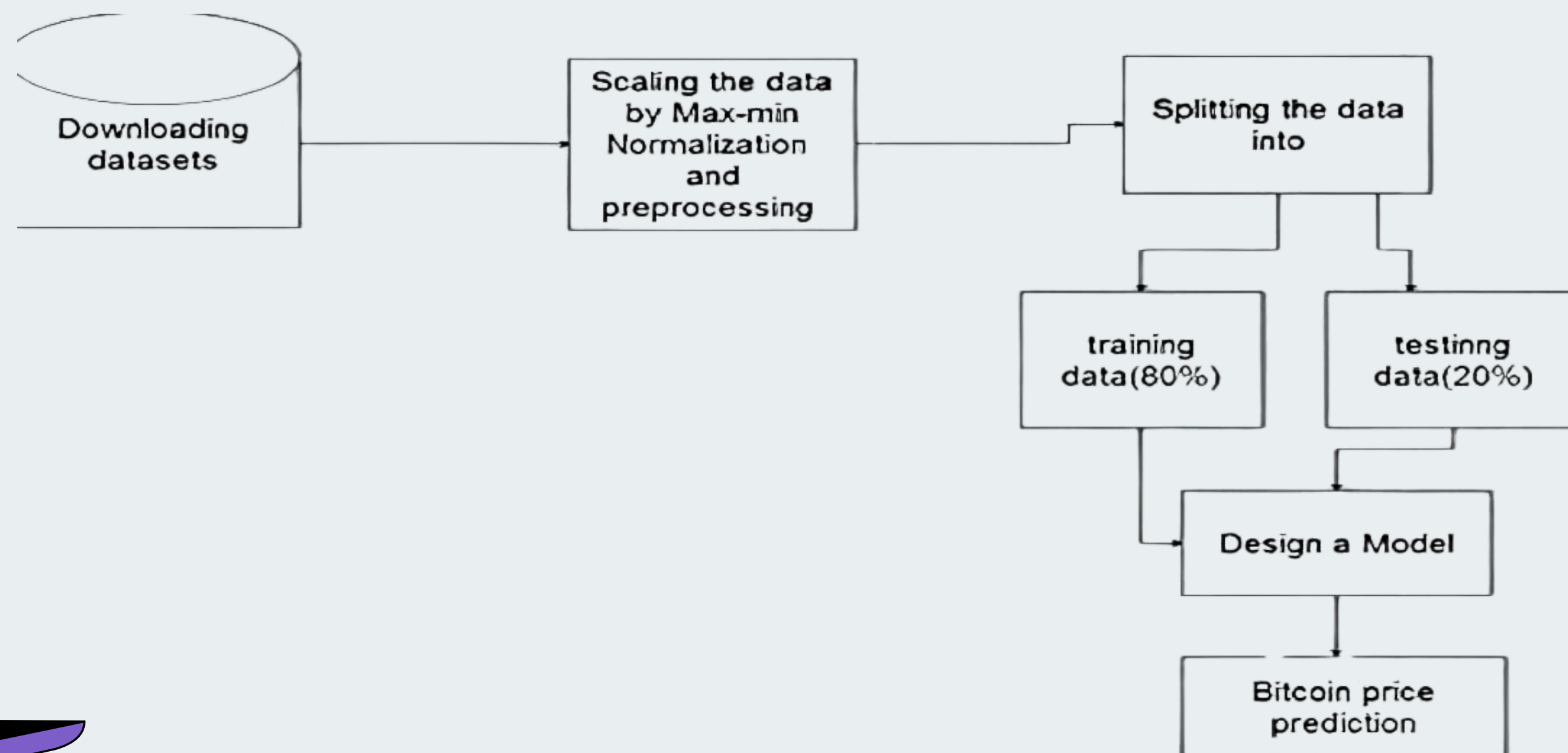
Bitcoin is a decentralized digital currency that was introduced in 2009 by an anonymous creator or group using the name Satoshi Nakamoto. It is powered by a technology called blockchain, which is a public ledger that records all Bitcoin transactions in a secure and transparent manner. One of Bitcoin's most defining features is its limited supply—only 21 million bitcoins will ever exist, making it a scarce digital asset often compared to gold.

Key terms of Bitcoin

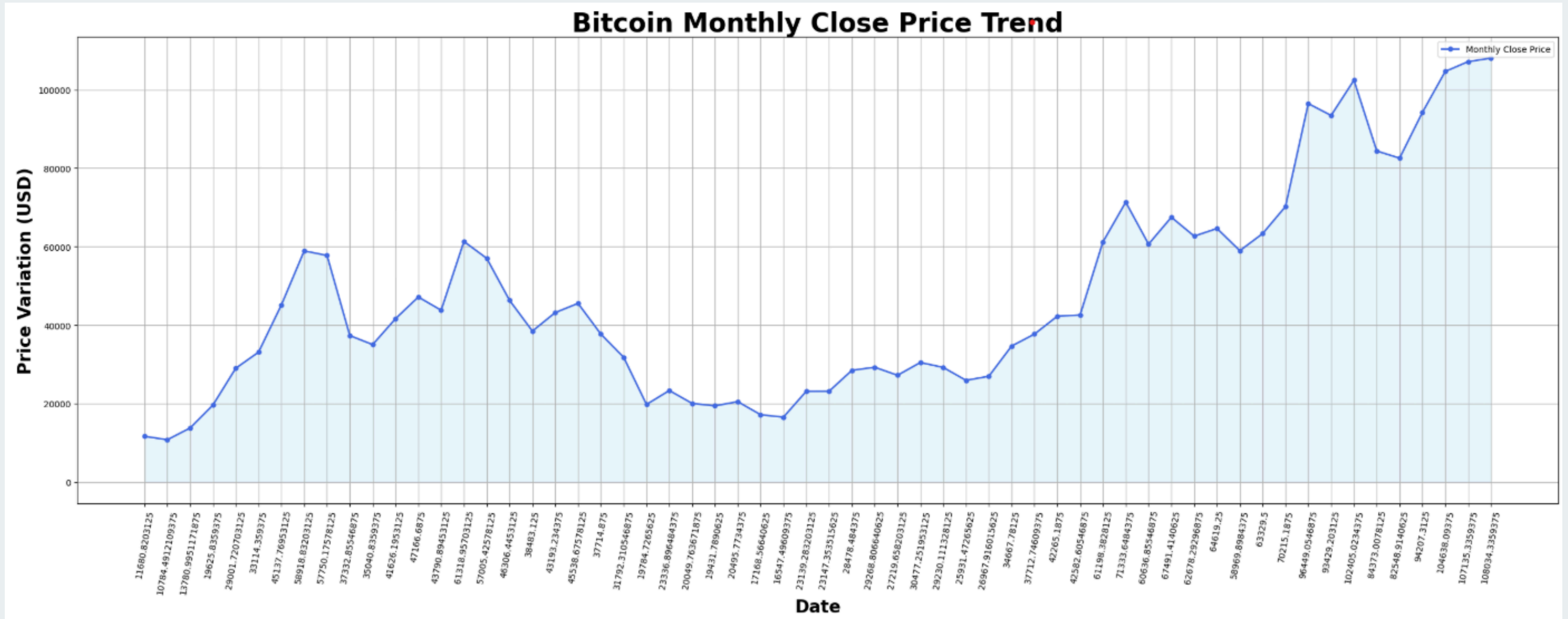
- Open Price: Price at the start of the day.
- Close Price: Final price of the day.
- High/Low: Daily highest and lowest values.
- Volume: Amount of Bitcoin traded.
- Change %: Day-over-day percentage change.



System ML Diagram

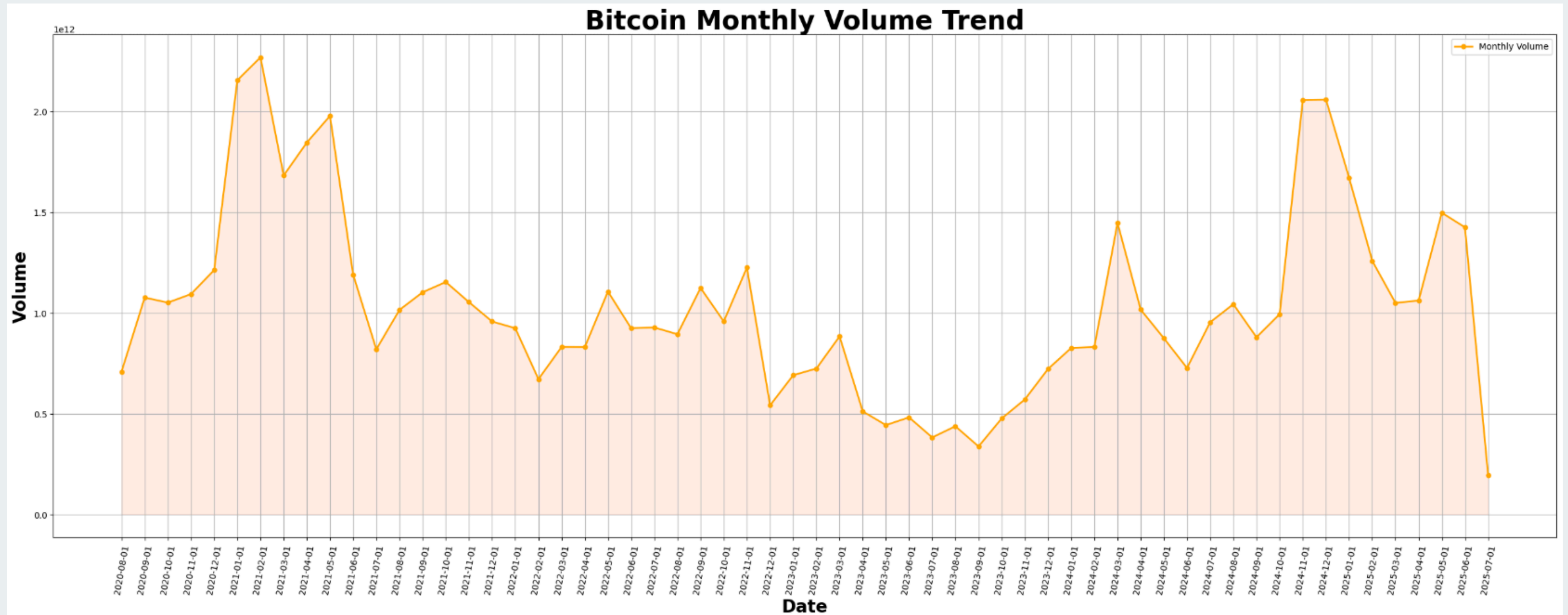


Bitcoin Monthly Close Price



Displays the monthly closing prices of Bitcoin over 5 years. The graph highlights Bitcoin's price growth, fluctuations, and key turning points. This graph shows how Bitcoin's value has changed month by month

Bitcoin Monthly Volume Trends



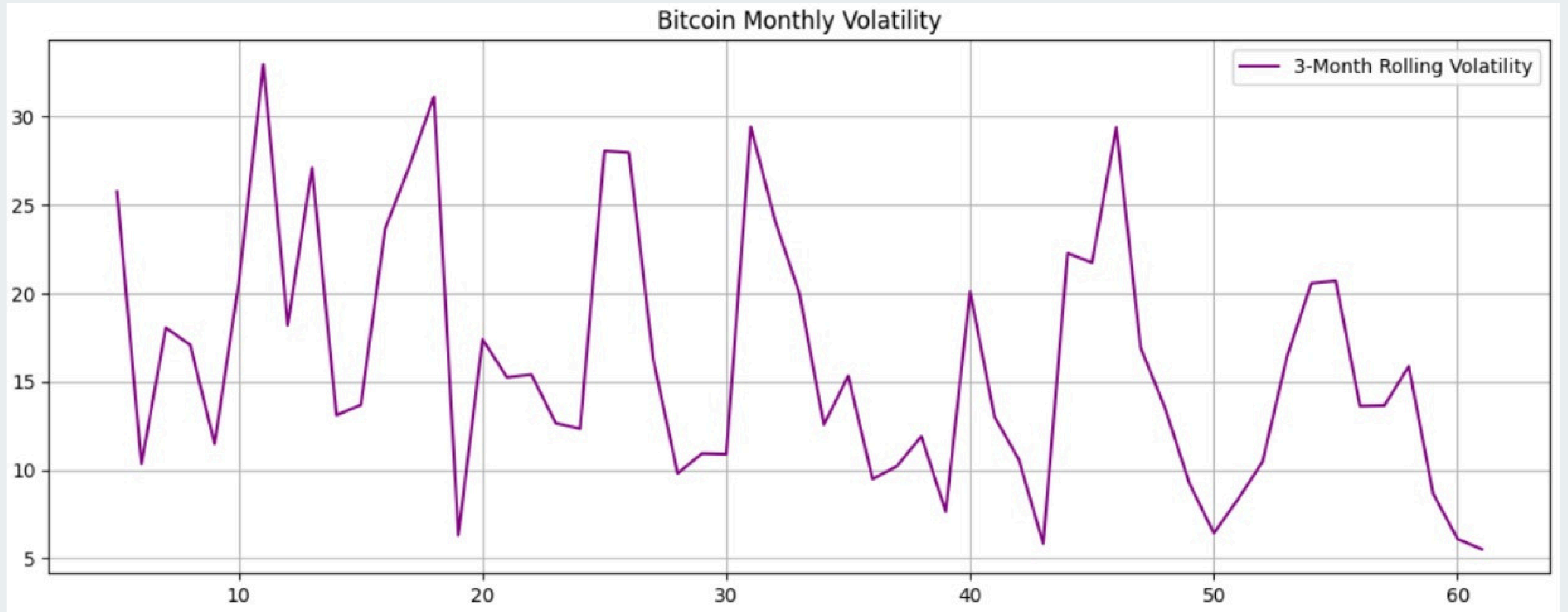
Illustrates the monthly trading volume of Bitcoin. Peaks indicate high buying or selling activity; troughs show reduced market interest. Higher volume usually means increased trader activity

Bitcoin Monthly % Returns



Shows monthly percentage returns, highlighting Bitcoin's high volatility. Both large gains and losses are visible. This helps us understand how much Bitcoin's price changes in percentage terms every month-useful to measure profitability and risk.

Bitcoin Monthly Volatility



Depicts Bitcoin's price volatility using a 3-month rolling average. Volatility measures how sharply prices move. High volatility means riskier but potentially more profitable trades, while low volatility indicates stability.

Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous values. It assumes a linear relationship between the input variables and the output variable.

Why it's used:

- To predict a value (e.g., house price, stock price)
- It's a good starting point for regression tasks due to simplicity



Linear Regression Prediction

- **MSE--> Closer to 0 is better**

(The squared error is quite large because price values are in thousands not a problem on its own, but shows big deviations in some predictions.)

- **MAE--> Closer to 0 is better**

(On average, your model's predictions are off by \$3,700, which is substantial given Bitcoin's price range)

- **R^2 --> 1 = perfect, 0 = bad, negative = worse than baseline**

(Your model explains 92.8% of the variance very strong for a simple linear model)

```
MSE: 19148003.73587479
MAE: 3700.7346745901646
R2: 0.9280581186188158
```

Random Forest

Random Forest is a powerful and flexible machine learning algorithm that combines the results of many decision trees to make a prediction. It's part of the ensemble learning family.

Why it's used:

- Handles nonlinear and complex data much better than linear models
- Works well with both classification and regression
- Automatically captures interactions between features



Random Forest Prediction

- **MSE--> Closer to 0 is better**

(The squared error is quite large because price values are in thousands not a problem on its own, but shows big deviations in some predictions.)

- **MAE--> Closer to 0 is better**

(On average, your model's predictions are off by \$32,755 , which is substantial given Bitcoin's price range)

- **R^2 --> 1 = perfect, 0 = bad, negative = worse than baseline**

(Your model gives negative value which means it's the worse result for this model)

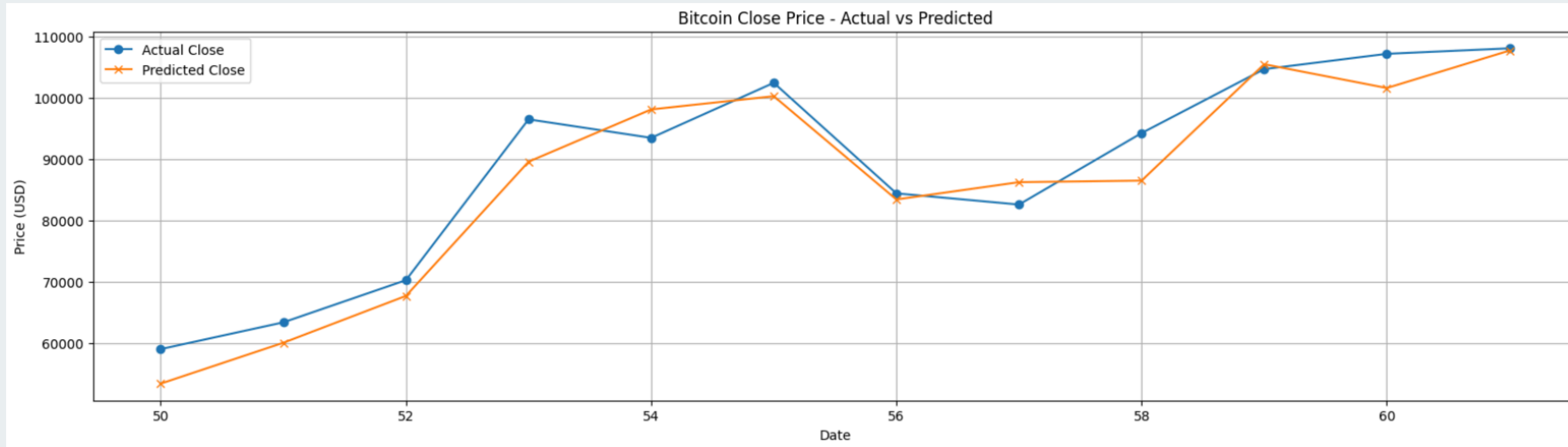
R^2 Score: -1.9200433011500464

MAE: 23755.034746093752

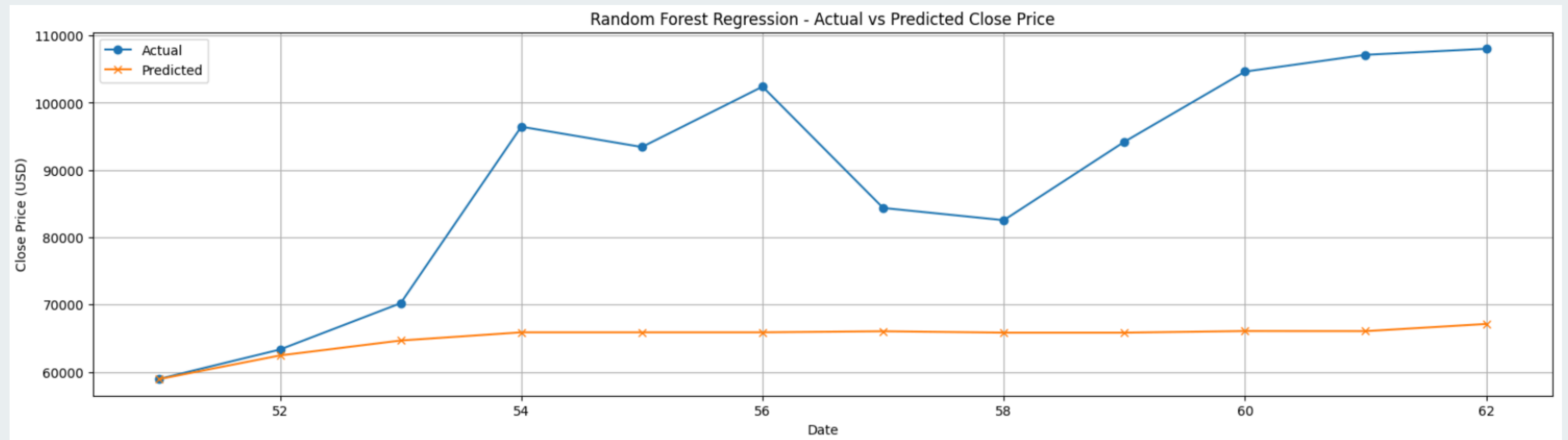
MSE: 777196800.6102332

Actual vs Predicted Close Price Prediction

Linear
Regression



Random
Forest



Time to Invest

Thank You For Your Attention

